

SAT-SA: supervisory assessment architecture

Smart India Hackathon · NCIIPC problem statement · Local offline prototype

Purpose. Help examiners identify which Critical Sector Entities (CSEs) and submitted SOC records warrant human review. The system examines periodic evidence. It is not a SOC, SIEM, continuous monitoring platform, or automated compliance decision maker.

Submitted evidence → **normalization** → **analytics** → **gaps and anomalies** → **review priority** → **human examination** → **auditable decision**

Components and boundaries

Layer	Implementation and responsibility
Presentation	Next.js/React/TypeScript. Overview, entity and signal views, source drawer, import, comparison, report, review, validation and audit.
API and workflow	FastAPI/Pydantic versioned API. Validates submissions, owns identity and roles, persists jobs, serves paginated evidence and reports.
Analytics	Independent Python modules for eight supervisory families, evidence gaps, peer statistics, prioritization and validation. Versioned configuration and reproducible run records.
Evidence	Immutable entity-scoped Parquet artifacts, queried through DuckDB. Ground truth is stored in a separate tree and excluded from ordinary evidence APIs.
Metadata	DuckDB database for dataset registrations, runs, jobs, reviews and append-only audit events. One API writer process in the prototype.

Submission and analysis

CSV/JSON exports are previewed, mapped, validated and committed as immutable dataset versions. Every normalized record retains source file, source record, ingestion batch and transformation version. Relationships use both CSE ID and local record ID. The engine reads submitted alert, case, investigation, escalation and asset records. No raw logs are required for the demo.

Rules identify short high-severity closures, weak investigation trails, missing escalation evidence, recurring activity, clustered closures, workload concentration, missing fields and monitoring expectations. Peer comparison requires a matched cohort. Missing inventory or insufficient peers produces an explicit unavailable result. Thresholds are versioned and included in each run.

Safety of supervisory interpretation

Each indicator provides observation, expected behavior and its basis, deviation, evidence count, completeness, method and source references. It is a review prompt. Alert absence cannot prove monitoring failure. Only an examiner can record a confirmed concern. Reviews append actor, time, outcome and note without changing the analytical result.

Deployment, validation and limits

Architecture handout · page 2 of 2

Local deployment

`docker compose up --build --wait` starts web and API containers with a persistent local volume. The API container runs one worker because embedded DuckDB has a single metadata writer. Runtime images bundle application assets and dependencies. The tested default path uses no cloud service, Internet request, external model, remote font or CDN. An optional external Mistral drafting mode and optional Render synthetic demo are explicitly outside that default offline path.

Examiner workflow

Step	What the examiner sees
1. Overview	Actual submitted volume and attention distribution across CSEs.
2. Entity and indicator	Sector, cohort, sample size, completeness, observations and rule rationale.
3. Evidence	Paginated source references resolve to the normalized record and provenance.
4. Human review	Recorded outcome and note, preserved in audit history.
5. Report and validation	Printable run report, separate synthetic-label evaluation and human-review counts.

Cross-period comparison

Two completed, non-overlapping runs are compared for the same CSE and declared cohort. Alert and case counts are converted to 30-day rates, so unequal assessment windows are not compared as raw totals. Missing entities and cohort changes are explicitly unavailable. A rate change is descriptive; source records and submission completeness still require review.

Validation evidence and next gate

The seeded local demo has eight pseudonymous CSEs across two sectors, 6,816 alerts, 852 cases, 192 assets, investigation events and escalation records. Separately stored synthetic labels cover injected scenarios and a healthy control. Tests exercise deterministic generation, strict lifecycle validation, immutable storage, reference integrity, API workflow, UI path and offline containers. Synthetic detection metrics are not an expert-review benchmark.

Before production use: compare prioritized samples with blinded expert manual review, measure precision and recall at the supervisor's review budget, calibrate thresholds by sector, test larger workloads and concurrent access, and complete security accreditation in the receiving environment.

Known prototype bounds

- Import limits: 2 MB and 10,000 rows per file; analytical loading is capped at 100,000 records. Million-record readiness is not claimed.
- DuckDB metadata is single-writer. PostgreSQL remains a documented migration path for stronger concurrency.
- Database exports work through CSV/JSON; the internal REST adapter is disabled unless explicitly configured.
- The demo is synthetic. No measured NCIIPC productivity or operational assurance outcome is claimed.

Source: SAT-SA repository docs/architecture.md, analytics methodology, validation methodology and completion verification. This handout summarizes the approved architecture; the repository documents implementation details and deviations.